

# WEIYI HE

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## EDUCATION

### Michigan State University

*Ph.D. in Statistics & Probability*

*Ph.D. in Computer Science & Engineering*

Aug. 2024 – Now

East Lansing, MI

### University of Science and Technology of China (USTC)

*Bachelor of Science in Statistics (Outstanding Graduate Award)*

Sep. 2020 – Jul. 2024

Hefei, China

## WORK EXPERIENCE

### Articuler AI, Inc.

*Generative AI Development Intern (Remote)*

Jun. 2024 – Aug. 2024

San Francisco, CA

- Spearheaded the development of the **Search Agent** product — a scalable Python pipeline that automates professional background research for end users, from data collection to structured insight generation. (Deployed to production)
- Designed multi-step LLM workflows to parse user profiles, generate targeted search queries, extract and validate information from industry and news sources via Google Search API and custom web-scraping logic, improving retrieval accuracy and end-user satisfaction.

### Elven Technologies Pte. Ltd.

*Data Engineering Intern (Remote)*

Jan. 2024 – May. 2024

Singapore

- Built Python API integrations with major cryptocurrency exchanges (e.g., Coinbase, Binance, OKX) to automate transaction extraction and synchronization for Elven's Web3 financial reporting platform.
- Parsed and formatted transaction logs, enabling reliable downstream reporting and visualization in the platform.

## RESEARCH AND OTHER PROJECTS

### ReFT-based latent adversarial training on low-dimensional manifolds

Dec. 2025 – Now

- Developed a defense strategy combining Low-rank Representation Fine-tuning (LoReFT) with Latent Adversarial Training (LAT); significantly mitigated Greedy Coordinate Gradient (GCG) attacks by implementing a multi-position intervention mechanism that prevents adversarial signal accumulation across causal sequences.
- Engineered a **Circuit-Aware** training framework utilizing Attribution Patching to isolate critical model circuits; optimized the inner-loop optimization by dynamically pruning non-essential gradient paths, achieving a 30% reduction in backward pass FLOPs while maintaining robust defensive performance.

### Impact of Positional Encoding on Transformer Generalization

May. 2025 – Oct. 2025

- Presented the first **generalization and robustness theoretical analysis** of single-layer Transformers with trainable positional encoding (PE) under the in-context regression, bridging theoretical understanding with empirical behavior.
- Derived both clean and adversarial Rademacher complexity bounds, demonstrating that trainable PE systematically enlarges the generalization gap and increases the model's sensitivity and vulnerability under adversarial perturbations.

### Privacy Risks in LLM Agent Memory

Oct. 2024 – Feb. 2025

- Studied how LLM agents storing user-agent interactions can expose private information through memory modules.
- Proposed **MEXTRA (Memory EXTRAction Attack)**, a black-box framework combining targeted attacking prompts with automated prompt generation under varying attacker knowledge levels.
- Analyzed architectural and deployment factors influencing memory vulnerability; demonstrated MEXTRA succeeded in extracting at least one private item in **83%–90%** of targeted trials across two real agents, highlighting the urgent need for effective memory safeguards in LLM agent designs.

## PUBLICATIONS & MANUSCRIPTS

- **Weiyi He**, Yue Xing. "Impact of Positional Encoding: Clean and Adversarial Rademacher Complexity for Transformers Under In-Context Regression." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.
- Bo Wang, **Weiyi He**, Shenglai Zeng, Zhen Xiang, Yue Xing, Jiliang Tang, and Pengfei He. "Unveiling Privacy Risks in LLM Agent Memory." *Proceedings of the Association for Computational Linguistics (ACL)*, 2025.

## TECHNICAL SKILLS

**Programming:** Python, R, C, Sklearn, Numpy, Pandas, PyTorch, Tensorflow, Huggingface

**Tools:** Git, Linux, Jupyter, VS Code

**Skills:** Large Language Models, Trustworthy AI, Data Analysis, Statistical Learning Theory